1 Theorem 16.7

Let $f : X \to Y$ be continuous and let $E \subset X$ be connected. Then $f(E) \subset Y$ is connected.

Proof by Contradiction. Suppose $E \subset X$ is connected but suppose for contradiction that $f(E) \subset Y$ is not connected. We may write $f(E) = A \cup B$ with open, disjoine, nonempty sets A and B in $(f(E), d_Y)$. By properties of images and preimages, we have $E \subset f^{-1}(f(E))$.

$$f^{-1}(f(E)) = f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B) \tag{1}$$

Therefore, $E = (f^{-1}(A) \cap E) \cup (f^{-1}(B) \cap E)$. Let $C = f^{-1}(A) \cap E$ and $D = f^{-1}(B) \cap E$.

<u>C and D are disjoint.</u> If they weren't disjoint, then there would be x such that $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$. That would mean that $f(x) \in A$ and $f(x) \in B$, which would not be possible since A and B are disjoint.

<u>C and D are nonempty.</u> A and B are nonempty. If $C = (f^{-1}(A) \cap E)$ were empty, then E would be a subset of $f^{-1}(B)$. This would imply $f(E) \subset B$, but we know $f(E) \subset B$, which would mean $f(E) = B$. That would mean $A = \varnothing$, but we already established this is false.

<u>C and D are open in (E, d_X).</u> Since A and B are open, we know that for U and V open in Y, $A = U \cap f(E)$ and $B = V \cap f(E)$. We can therefore say the following:

$$f^{-1}(A) = f^{-1}(U \cap f(E)) = f^{-1}(U) \cap f^{-1}(f(E)) \tag{2}$$

$$f^{-1}(B) = f^{-1}(V \cap f(E)) = f^{-1}(V) \cap f^{-1}(f(E)) \tag{3}$$

$$f^{-1}(A) \cap E = f^{-1}(U) \cap f^{-1}(f(E)) \cap E = f^{-1}(U) \cap E \tag{4}$$

$$f^{-1}(B) \cap E = f^{-1}(V) \cap f^{-1}(f(E)) \cap E = f^{-1}(V) \cap E \tag{5}$$

Since by continuity $f^{-1}(U)$ and $f^{-1}(V)$ are open in (X, d_X), $f^{-1}(A) \cap E$ is open in (E, d_X) and $f^{-1}(B) \cap E$ is open in (E, d_X), so C and D are open in (E, d_X).

Therefore, $E = C \cup D$ for open, disjoint, nonempty C and D. Therefore, E is separated, which is a contradiction. Ergo, $f(E)$ is connected. □

2 Theorem 17.7

Let $f : X \to Y$ be continuous between metric spaces. Let X be compact. Then, f is uniformly continuous.

Proof. Let $\varepsilon > 0$. For all $p \in X$, there exists $\delta_p > 0$ (depending on p) such that if $d_X(p,q) < \delta_p$, then $d_Y(f(p), f(q)) < \frac{\varepsilon}{2}$. Therefore, the open sets $\left\{ B_{\frac{\delta_p}{2}}(p) \right\}$ cover X. Since X is compact, there is a finite subcover $\left\{ B_{\frac{\delta_{p_k}}{2}}(p_k) \right\}_{k=1}^{n}$.

$$X \subset \left\{ B_{\frac{\delta_{p_k}}{2}}(p_k) \right\}_{k=1}^{n} \tag{$\star$}$$

Let $\delta = \min_{k \in \mathbb{N} \leq n} \left\{ \frac{\delta_{p_k}}{2} \right\} > 0$. Choose any $x, y \in X$ with $d_X(x,y) < \delta$. Since $(\star)$ is true, there exists j where $1 \leq j \leq n$ and $x \in B_{\frac{\delta_{p_j}}{2}}(p_j)$. Therefore, we can say things about $d(x, p_j)$ and $d(y, p_j)$.

$$d_X(x, p_j) < \frac{\delta_{p_j}}{2} < \delta_{p_j} \tag{6}$$

$$d_X(y, p_j) \leq d_X(x,y) + d_X(x, p_j) \leq \frac{\delta_{p_j}}{2} + \frac{\delta_{p_j}}{2} = \delta_{p_j} \tag{7}$$

Since f is continuous,

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(p_j)) + d_Y(f(p_j), f(y)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \tag{8}$$

Therefore, f is uniformly continuous. $\square$